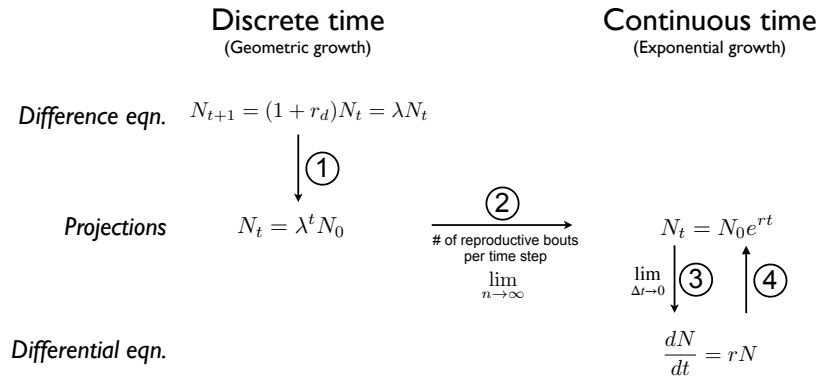


Lecture 2 - Summary

The purpose of last lecture was to show connections between alternative descriptions of population growth. So, to summarize:



① We started with $N_{t+1} = N_t + B + D + I + E$, then substituted $E = I$, $B = bN_t$ and $D = dN_t$, where b and d were per capita rates. Using $r_d = b - d$ and $\lambda = 1 + r_d$, we then simplified down to $N_{t+1} = (1 + r_d)N_t = \lambda N_t$. Thus, for example,

$$N_{t+2} = \lambda N_{t+1} = \lambda(\underbrace{\lambda N_t}_{N_{t+1}}) = \lambda^2 N_t,$$

such that more generally we can write $N_t = \lambda^t N_0$ for any arbitrary t time-steps into the future.

② From the discrete to the continuous: Let's say that in $t=1$ year an annually reproducing (and dying) population grows by a factor of λ . From that one reproductive event we'd have:

1 event per time-step:

$$N_1 = \lambda N_0 = (1 + r_d)N_0$$

Now, if we had 2 reproductive events per year but achieved the same total amount of population growth over the year, we'd write

2 events per time-step:

$$N_1 = \left(1 + \frac{r_d}{2}\right)^2 N_0$$

By extension, if we had an n reproductive events per year but achieved the same total amount of population growth over the year, we'd write:

n events per time-step:

$$N_1 = \left(1 + \frac{r_d}{n}\right)^n N_0$$

Now, λ is nothing more than the discrete population growth factor (how much the population size changed over one year), so let's divide both sides by N_0 to isolate λ :

$$\lambda = \frac{N_1}{N_0} = \left(1 + \frac{r_d}{n}\right)^n$$

We won't prove it, but by demonstration (see next page):

$$\lambda = \lim_{n \rightarrow \infty} \left(1 + \frac{r_d}{n}\right)^n = e^r$$

Therefore, we have

$$\lambda = \left(1 + \frac{r_d}{1}\right) = e^r$$

and we can relate (convert between) discrete-time and continuous-time population growth:

$$N_t = N_0 \lambda^t = N_0 e^{rt}$$

How does Euler's number, e , emerge?

Let's let $n = 1$, $N_0 = 1$, $r_d = 1$

The true of $e = \exp(1) = 2.71828\dots$, but we'll estimate it as $\lambda = \frac{N_1}{N_0}$ for ever increasing values of n :

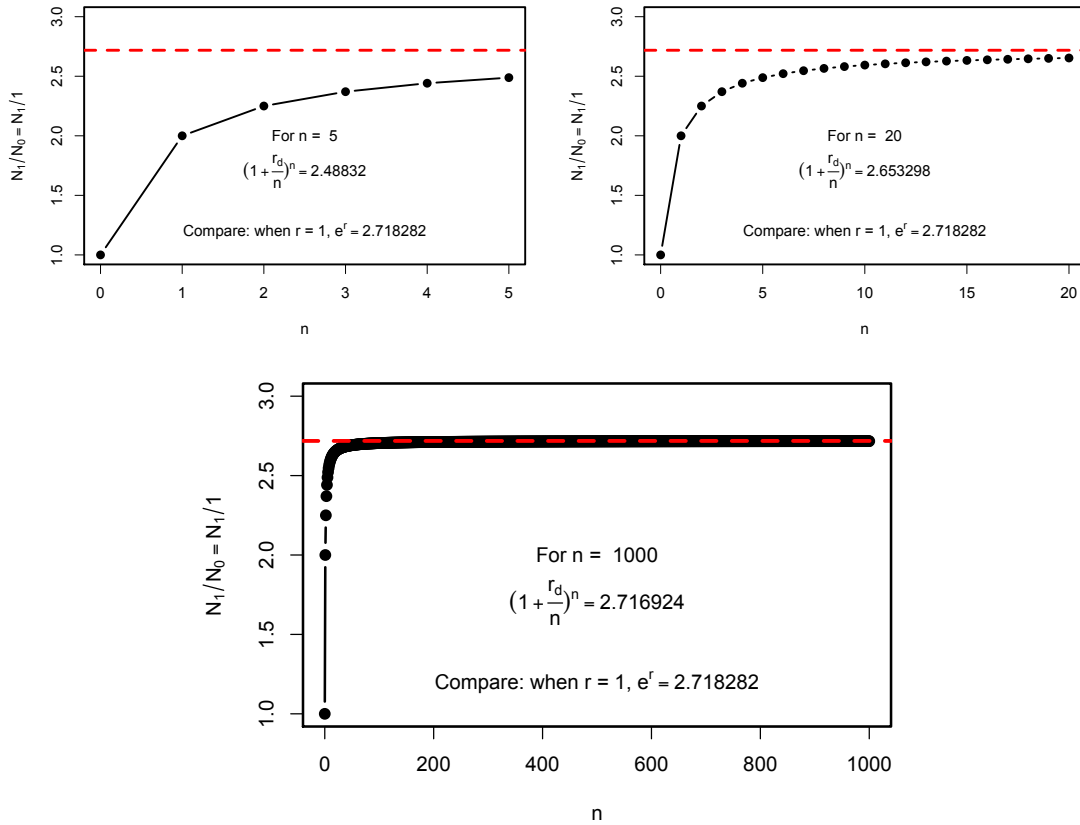
$$n = 1 \Rightarrow \frac{N_1}{N_0} = \left(1 + \frac{1}{1}\right)^1 = 2$$

$$n = 2 \Rightarrow \frac{N_1}{N_0} = \left(1 + \frac{1}{2}\right)^2 = 2.25$$

$$n = 3 \Rightarrow \frac{N_1}{N_0} = \left(1 + \frac{1}{3}\right)^3 = 2.30707\dots$$

$$n \rightarrow \infty \Rightarrow \frac{N_1}{N_0} = \left(1 + \frac{1}{n}\right)^n = 2.71828\dots = e$$

In graphical form:



Defining the natural-log as the 'anti-exponential' — $\ln(e^x) = x$ — we also have:

$$\ln(\lambda) = \ln(1 + r_d) = r$$

All three (λ , r_d , and r) reflect a *per capita growth rate* for either geometric or exponential growth:

$$r_d = b_d - d_d$$

discrete time

$$r = b - d$$

continuous time

③ How to we get instantaneous population-level growth rate from projection equation, $N_0 e^{rt}$?
That is, how do we show that:

$$\lim_{\Delta t \rightarrow 0} \left(\frac{\Delta N_t}{\Delta t} \right) = \frac{dN_t}{dt}$$

Need to take the derivative of $N_0 e^{rt}$ with respect to time t .

Use Chain Rule:

$$\frac{d(XY)}{dt} = \frac{d(X)}{dt} \cdot Y + X \cdot \frac{d(Y)}{dt}$$

(The derivative of a product is the sum of the product of the derivative of each term times the other term.)

Thus:

$$\frac{d(N_0 \cdot e^{rt})}{dt} = \frac{d(N_0)}{dt} \cdot (e^r)^t + N_0 \cdot \frac{d((e^r)^t)}{dt}$$

Note:

Derivative of a constant = 0

Derivative of $a^x = \ln(a) \cdot a^x$.

Thus:

$$\begin{aligned} \frac{d(N_0 e^{rt})}{dt} &= 0 \cdot (e^r)^t + \ln(e^r) \cdot (e^r)^t \cdot N_0 \\ &= r \cdot (e^r)^t \cdot N_0 \\ &= r \cdot e^{rt} \cdot N_0 \\ &= r \cdot N_0 e^{rt} \end{aligned}$$

Since $N_t = N_0 e^{rt}$ for any time t ...

$$= r N_t = \frac{dN_t}{dt} \quad \text{...then drop } t \text{ subscript}$$

④ Could also go in opposite direction from $\frac{dN}{dt} \rightarrow N_0 e^{rt}$:

$$\begin{aligned} \frac{dN}{dt} &= rN \\ \frac{1}{N} \frac{dN}{dt} &= r \\ \int_0^T \frac{1}{N} \frac{dN}{dt} dt &= \int_0^T r dt \quad (\text{Think of } T \text{ as a constant, and } t \text{ in } dt \text{ as a variable}) \\ \int_0^T \frac{1}{N} \frac{dN}{dt} dt &= rt \Big|_0^T = r \cdot T - r \cdot 0 = rT \end{aligned}$$

Using $\int \frac{1}{x} dx = \ln(x)$. That is, $\int_{x_0}^{x_T} \frac{1}{x} dx = \ln(x_T) - \ln(x_0)$, we have that

$$\begin{aligned} \ln(N_T) - \ln(N_0) &= rT \\ \ln\left(\frac{N_T}{N_0}\right) &= rT \\ \frac{N_T}{N_0} &= e^{rT} \\ N_T &= N_0 e^{rT} \end{aligned}$$